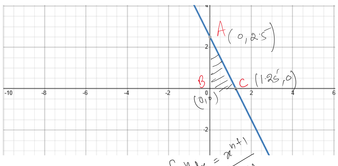

$4x = -2y + 5$

$\frac{0}{2}, \frac{A}{2}$

$(0, 2.5)$

$2y = -4x + 5$

$\boxed{2y = 5 - 4x}$

$\frac{1}{2} \times b \times h$

$= \frac{1}{2} \times 1.25 \times 2.5$

$= \frac{3.125}{2}$

$= 1.56$ squarks

$\int x^n dx = \frac{x^{n+1}}{n+1}$

only those points to be taken where points will be satisfied

A $(0, 2.5)$ $\quad 2y = 5 - 4x$

$y = \frac{5}{2} - \frac{4x}{2} = y = 2.5 - 2x$

$2.5 = 2.5 - 2 \times (0)$ ✓

B $(0,0)$ $\quad 2y = 5 - 4x$

$0 = 5$ ✗

$(0, 2.5) \rightarrow (1.25, 0)$

C $(1.25, 0)$ $\quad y = 2.5 - 2x$

$2.5 - 2(1.25)$ ✓

$0 = 2.5 - 2.5$

$\int_0^{1.25} \rightarrow x$

$\int_0^{2.5} \rightarrow y$

$$\int_0^{1.25} y\,dx = \int_0^{1.25} (2.5 - 2x)\,dx$$

$$\int_0^{1.25} 2.5\,dx - \int_0^{1.25} 2x\,dx$$

$$= 2.5 \left| x \right|_0^{1.25} - \left| \frac{x^2}{2} \right|_0^{1.25} = \left| 1.25^2 - 0^2 \right|$$

$$= 2.5 \left| 1.25 - 0 \right| - (1.25)^2$$

$$2.5 \times 1.25 - 1.56$$

$$= 3.125 - 1.56$$

$$= 1.565$$

Note:

linear $\rightarrow$ equ $\rightarrow$ formation $\rightarrow$ Q

Q $\rightarrow$ fn $\rightarrow$ cubic $\rightarrow$ Bi Quadratic

Eg: $x^2 = 4y - 5 \Rightarrow$

$x^3 = 4y$

Even fn: $y = |x|, x^2, x^4, x^6 \ldots\ldots x^{2n}$

odd fn $y = x^1, x^3, x^5, x^7 \ldots\ldots x^{2n-1}$

Homogenous

$x^{1.25}$

what is logorithm = log ?

$10^{\boxed{3}} = 1000$ (number)

base, power

$2^{\boxed{3}} = 8$ Number

base

$\log_2 8 =$ power 3

$\boxed{\log_{10} 1000 = 3}$

$\log_{\text{base } e} x = n$

e^x

increasing

$\log(x)$ $\quad x \in R$

$-\infty$

$y = e^x$ strictly increasing

$1, \infty$